ADHD Behavioral Prediction using Machine Learning and Explainable AI

A reproducible computational approach for ADHD classification using the ADHD-200 Consortium phenotypic dataset

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Project Overview

This project investigates whether machine learning models can identify patterns associated with Attention-Deficit/Hyperactivity Disorder (ADHD) using clinical phenotypic data.

The project consists of two interconnected studies:

Study 1 — Predictive Modeling

Development and evaluation of machine learning models for ADHD classification.

Study 2 — Explainable Artificial Intelligence

Analysis of model decisions to determine which features contribute most strongly to predictions.

Repository

ADHD-Behavioral-Prediction-AI

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Background and Motivation

Background

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by persistent patterns of inattention and/or hyperactivity-impulsivity that can significantly affect daily functioning.

ADHD diagnosis is primarily based on clinical assessment, behavioral evaluation, and symptom reporting. Because ADHD presents with considerable variation between individuals, identifying consistent patterns across patients remains an important research challenge.

Motivation

Machine learning provides a framework for analyzing complex clinical datasets and identifying relationships between multiple characteristics that may not be obvious through traditional statistical approaches.

This project explores whether artificial intelligence methods can:

- Identify patterns associated with ADHD diagnosis
- Combine multiple clinical and cognitive measurements
- Provide predictive models based on phenotypic information
- Reveal which characteristics contribute most strongly to predictions

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Importance of Explainability

In healthcare applications, predictive accuracy alone is not sufficient.

Machine learning models can achieve high performance while remaining difficult to interpret. Understanding **why** a model produces a specific prediction is essential for:

- Scientific validation
- Clinical trust
- Responsible use of artificial intelligence

For this reason, the project includes an explainable AI analysis to investigate the factors driving model predictions.

Overall Goal

The goal of this work is not to replace clinical diagnosis, but to investigate whether machine learning can discover meaningful patterns within existing ADHD-related data and whether these patterns can be interpreted in a scientifically meaningful way.

Research Questions and Project Structure

Research Objective

The main objective of this project is to investigate whether machine learning models can identify ADHD-related patterns from clinical phenotypic data and whether these predictions can be interpreted using explainable artificial intelligence methods.

The project is divided into two studies.

Study 1 — Predictive Modeling

Research Question

Can machine learning models predict ADHD diagnosis using phenotypic characteristics?

Objective

To develop and evaluate classification models that distinguish between participants with and without ADHD using available clinical and cognitive features.

The study investigates:

- Whether meaningful predictive patterns exist within phenotypic data
- Which machine learning algorithms perform best
- How accurately ADHD classification can be achieved

Approach

The predictive pipeline consists of:

1. Dataset acquisition
2. Data cleaning and preprocessing
3. Feature preparation
4. Machine learning model training
5. Performance evaluation
6. Selection of the best-performing model

Study 2 — Explainable Artificial Intelligence

Research Question

Which features contribute most strongly to machine learning predictions, and are these contributions clinically meaningful?

Objective

To understand the decision-making process of the final predictive model using explainable AI techniques.

The study investigates:

- Which variables influence predictions
- The relative importance of different features
- Whether model behavior aligns with known ADHD characteristics

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Project Structure

The complete workflow:

ADHD-200 Phenotypic Dataset

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Data Processing

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Machine Learning Models

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Study 1:
ADHD Prediction

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Best Model Selection

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Study 2:
SHAP Explainability

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Interpretation of Model Behavior

Central Research Question

Can artificial intelligence predict ADHD-related patterns from clinical data, and can these predictions be understood rather than treated as a black box?

Dataset Description: ADHD-200 Consortium Phenotypic Dataset

Dataset Overview

This project uses the **ADHD-200 Consortium phenotypic dataset**, a publicly available dataset created to support research into ADHD classification and neurodevelopmental differences.

The ADHD-200 Consortium collected clinical, behavioral, and imaging-related information from multiple international research sites.

For this project, the phenotypic component of the dataset was used.

Dataset Characteristics

Initial dataset:

- **973 participants**
- 19 available variables
- Multiple research sites represented

After data cleaning and preparation:

- **797 participants retained**
- Binary ADHD classification target created

Prediction Target

The original diagnosis variable was transformed into a binary classification problem:

Class	Meaning
0	No ADHD diagnosis
1	ADHD diagnosis

Final class distribution:

Class	Participants
No ADHD	585
ADHD	212

Available Features

The dataset contained several categories of information:

Demographic Features

- Age
- Gender
- Handedness
- Research site

Cognitive Measurements

- Verbal IQ
- Performance IQ
- Full Scale IQ measurements

ADHD-Related Clinical Features

- ADHD Index
- Inattentive symptom score
- Hyperactive/Impulsive symptom score
- ADHD measurement scores

Dataset Selection Rationale

Phenotypic data was selected because it provides a clinically meaningful starting point for machine learning analysis.

Unlike raw biological signals, these variables represent measurable characteristics already used in ADHD research and allow investigation of whether AI can reproduce clinically relevant patterns.

Dataset Limitation

The dataset represents research participants and should not be interpreted as a clinical diagnostic population.

The resulting models are intended for:

- Research exploration
- Pattern identification
- Machine learning methodology evaluation

They are **not clinical diagnostic tools**.

Data Processing and Machine Learning Pipeline

Data Preparation

Before training machine learning models, the raw ADHD-200 phenotypic dataset required preprocessing to create a clean and consistent analysis dataset.

The preprocessing pipeline focused on:

- Handling missing values
- Converting variables into machine-readable formats
- Creating a binary ADHD classification target
- Preparing numerical and categorical features for model training

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Data Processing Workflow

Raw ADHD-200 Dataset

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Data Inspection

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Missing Value Handling

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Feature Cleaning

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Diagnosis Label Transformation

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Machine Learning Dataset

Data Cleaning Steps

1. Dataset Validation

The dataset was inspected for:

- Available variables
- Missing values
- Data consistency
- Diagnosis categories

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2. Target Variable Preparation

The original diagnosis field contained multiple categories.

For supervised classification, diagnosis labels were transformed into:

- **0 → No ADHD**
- **1 → ADHD**

This created a binary classification problem suitable for machine learning.

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3. Feature Processing

The selected input features included:

Numerical Variables

- Age
- Verbal IQ
- Performance IQ
- Full4 IQ

- ADHD Index
- Inattentive score
- Hyper/Impulsive score
- ADHD measurement score

Categorical Variables

- Gender
- Handedness

Categorical variables were encoded into numerical representations for compatibility with machine learning algorithms.

Final Machine Learning Dataset

After preprocessing:

- Participants: **797**
- Features: Clinical, demographic, and cognitive variables
- Target: Binary ADHD classification

The resulting dataset was used for all predictive modeling experiments.

Reproducibility

All preprocessing steps were implemented programmatically and stored within the project repository, allowing the complete pipeline to be reproduced from the original dataset.

Study 1: Machine Learning Prediction Approach

Study Objective

The first study investigates whether machine learning models can predict ADHD classification from phenotypic characteristics.

The central question was:

Can clinical, demographic, and cognitive features be combined to accurately distinguish participants with ADHD from those without ADHD?

Machine Learning Task

The problem was formulated as a **supervised binary classification task**.

Input

Participant characteristics:

- Demographic variables
- Cognitive measurements
- ADHD-related symptom scores

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Machine Learning Model

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Output

Prediction:

- 0 → No ADHD
- 1 → ADHD

Models Evaluated

Multiple algorithms were compared to determine which approach best captured patterns within the dataset.

1. Logistic Regression

A classical statistical classification model used as a baseline.

Advantages:

- Simple interpretation
- Strong baseline performance
- Assumes linear relationships

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2. Support Vector Machine (SVM)

A model capable of identifying complex decision boundaries between classes.

Advantages:

- Effective in high-dimensional spaces
- Can model nonlinear relationships using kernels

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3. Random Forest

An ensemble model based on multiple decision trees.

Advantages:

- Captures nonlinear relationships
- Handles mixed feature types
- Provides feature importance estimates

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4. XGBoost

An optimized gradient-boosting algorithm.

Advantages:

- Strong performance on structured/tabular data
- Captures complex feature interactions
- Regularization reduces overfitting risk

Evaluation Strategy

Models were evaluated using:

ROC-AUC

Measures how well the model separates ADHD and non-ADHD participants across different classification thresholds.

Higher values indicate better discrimination.

Accuracy

Measures the percentage of correctly classified participants.

Additional Metrics

Classification performance was also evaluated using:

- Precision
- Recall
- F1-score
- Confusion matrix

Model Selection Strategy

The best-performing model was selected based on:

1. Cross-validation ROC-AUC performance
2. Stability across validation folds
3. Final test performance

The selected model was then used for the explainable AI analysis in Study 2.

Study 1 Results: Model Benchmarking and Performance Comparison

Model Evaluation

Four machine learning algorithms were trained and compared using cross-validation to determine which approach provided the strongest ADHD classification performance.

The primary evaluation metric was **ROC-AUC**, which measures the ability of a model to distinguish between participants with and without ADHD.

Cross-Validation Results

Model	Mean ROC-AUC	Standard Deviation
XGBoost	**0.9698**	0.0120
Random Forest	0.9503	0.0274
Support Vector Machine	0.9018	0.0096
Logistic Regression	0.7372	0.0353

Best Performing Model

XGBoost

The XGBoost model achieved the highest predictive performance:

- Mean cross-validation ROC-AUC: **0.9698**
- Stable performance across validation folds
- Strong ability to separate ADHD and non-ADHD participants

This model was selected as the final predictive model for further analysis.

Final Model Evaluation

The final XGBoost model was evaluated on a held-out test set.

Performance:

Metric	Value
Accuracy	92.5%
ROC-AUC	93.6%

Confusion Matrix

	Predicted	
	No ADHD	ADHD
Actual		
No ADHD	111	6
ADHD	6	37

Interpretation

The model correctly classified the majority of participants while maintaining balanced performance between both groups.

The results demonstrate that:

- Phenotypic characteristics contain meaningful predictive information
- Nonlinear machine learning approaches outperform simpler linear models
- XGBoost provides a strong foundation for further explainability analysis

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Important Note

High predictive performance does not mean the model has discovered a new biological cause of ADHD.

The model identifies statistical patterns within the available clinical measurements.

Understanding which patterns drive these predictions is the focus of **Study 2: Explainable Artificial Intelligence**.

Feature Importance Analysis: Understanding What the Model Learns

This will connect Study 1 results to the need for SHAP explainability.

lets go

Why Analyze Feature Importance?

Machine learning models can achieve high predictive performance, but performance alone does not explain **which characteristics drive predictions**.

To better understand the model behavior, feature importance analysis was performed on the final tree-based models.

The goal was to identify which variables contributed most strongly to ADHD classification.

Feature Importance Results

The Random Forest model was analyzed to estimate the relative contribution of each feature.

Most Important Features

Rank	Feature	Relative Importance
1	Hyper/Impulsive symptoms	0.1937

2	ADHD Index	0.1910
3	Inattentive symptoms	0.1862
4	Age	0.1385
5	Full4 IQ	0.0853
6	Performance IQ	0.0492
7	Gender	0.0472
8	Verbal IQ	0.0437
9	ADHD Measure	0.0218

Main Observation

The strongest predictors were ADHD-related symptom measurements:

- ADHD Index
- Inattentive symptoms
- Hyperactive/Impulsive symptoms

These features represent the core behavioral dimensions associated with ADHD diagnosis.

Interpretation

The model appears to rely primarily on clinically relevant behavioral information rather than arbitrary demographic characteristics.

This suggests that:

- The predictive signal is driven mainly by symptom severity patterns
- The model captures relationships already recognized in ADHD assessment
- Demographic and cognitive variables provide additional but weaker information

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Limitation of Traditional Feature Importance

Although useful, standard feature importance methods have limitations:

- They show overall importance only
- They do not explain individual predictions
- They do not show whether a feature increases or decreases ADHD probability

For a deeper understanding of model decisions, an explainable AI approach was required.

Transition to Study 2

The next question becomes:

> **Why does the model make these predictions, and how much does each feature contribute to individual decisions?**

This motivates the second study:

Study 2 — Explainable Artificial Intelligence using SHAP

Study 2: Explainable Artificial Intelligence (SHAP Analysis)

Motivation

Machine learning models can achieve excellent predictive performance while remaining difficult to interpret.

This creates a challenge in healthcare-related applications:

> A model that predicts correctly is useful, but understanding **why** it predicts is essential for scientific interpretation and responsible use.

Therefore, the second study focuses on **Explainable Artificial Intelligence (XAI)**.

Research Question

Which features influence the model's predictions, and do these relationships correspond to meaningful ADHD-related characteristics?

Explainable AI Method: SHAP

This project uses **SHAP (SHapley Additive exPlanations)** to interpret the final machine learning model.

SHAP is based on cooperative game theory and estimates the contribution of each feature to a specific prediction.

In simple terms:

- Each feature receives a contribution value
- Positive values push predictions toward ADHD
- Negative values push predictions away from ADHD
- Larger absolute values indicate stronger influence

SHAP Analysis Workflow

Final XGBoost Model

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Processed Dataset

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SHAP Value Calculation

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Feature Contribution Analysis

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Interpretation of Model Decisions

Why SHAP Was Selected

SHAP provides several advantages:

Global Interpretation

Shows which features are generally most important across the entire dataset.

Individual Prediction Explanation

Shows why a specific participant received a specific prediction.

Model-Agnostic Framework

Can be applied to complex machine learning models without requiring a simple mathematical relationship.

Connection Between Study 1 and Study 2

Study 1 demonstrated:

> Machine learning can accurately classify ADHD from phenotypic information.

Study 2 investigates:

> Whether the model's decisions are based on clinically meaningful patterns.

Together, the two studies provide both:

- **Predictive performance**
- **Interpretability**

SHAP Results: What Drives ADHD Predictions?

SHAP Feature Contribution Analysis

After training the final predictive model, SHAP analysis was performed to investigate which variables contributed most strongly to model predictions.

Unlike traditional feature importance, SHAP provides information about **how much each feature contributes to model decisions**.

Top SHAP Features

Rank	Feature	Mean Absolute SHAP Value
1	ADHD Index	1.6816
2	Inattentive symptoms	0.9543
3	Full4 IQ	0.7540
4	Hyper/Impulsive symptoms	0.6431
5	Age	0.4999
6	Gender	0.4249
7	Verbal IQ	0.3525
8	Performance IQ	0.2854
9	ADHD Measure	0.2715

Main Findings

The SHAP analysis confirmed that the model relied primarily on ADHD-related behavioral measurements.

The strongest contributors were:

ADHD Index

The ADHD Index was the most influential feature, indicating that overall ADHD symptom severity strongly affected model predictions.

Inattentive Symptoms

Attention-related difficulties represented one of the strongest predictive patterns.

Hyperactive/Impulsive Symptoms

Hyperactivity and impulsivity measures also contributed substantially to classification decisions.

Comparison With Feature Importance

The SHAP results were consistent with the earlier feature importance analysis.

Both approaches identified:

1. ADHD symptom scores
2. Inattention measures
3. Hyperactivity/impulsivity measures

as the dominant predictive variables.

This consistency increases confidence that the model is learning meaningful patterns rather than relying on accidental relationships.

Scientific Interpretation

The model appears to reproduce clinically recognized ADHD dimensions:

- Inattention
- Hyperactivity/impulsivity

- Overall symptom burden

Rather than discovering an unknown diagnostic marker, the model is learning statistical relationships already represented in clinical measurements.

SHAP Visualization

The repository contains:

```
reports/
└── figures/
    └── shap_summary.png
```

This visualization summarizes:

- Feature importance
- Direction of influence
- Distribution of feature effects across participants

Scientific Interpretation and Discussion

Summary of Findings

This project investigated whether machine learning models could identify ADHD-related patterns from phenotypic clinical data and whether these predictions could be interpreted using explainable AI methods.

The results demonstrate that:

1. Machine learning models can successfully classify ADHD status from available phenotypic information.
2. Ensemble tree-based models, particularly XGBoost, achieved the strongest predictive performance.
3. Explainable AI analysis showed that predictions were mainly driven by clinically meaningful symptom-related variables.

Interpretation of the Predictive Model

The final model achieved high classification performance, suggesting that the ADHD-200 phenotypic dataset contains measurable patterns associated with ADHD diagnosis.

The model performed best when using nonlinear algorithms capable of detecting complex interactions between variables.

This indicates that:

- ADHD-related characteristics are not represented by a single variable
- Multiple clinical measurements contribute together to classification
- Machine learning can capture relationships between different aspects of patient data

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Interpretation of Explainable AI Results

The SHAP analysis demonstrated that the model relied primarily on:

- ADHD symptom severity measures
- Inattentive symptoms
- Hyperactive/impulsive symptoms

These findings align with established clinical descriptions of ADHD.

The model therefore appears to be learning patterns that correspond to known behavioral dimensions rather than relying mainly on unrelated demographic variables.

Important Scientific Perspective

The model should not be interpreted as discovering a new cause or biological marker of ADHD.

Instead, it demonstrates that:

> Machine learning can reproduce and quantify patterns present within existing clinical assessments.

The value of the approach is not replacing clinical evaluation, but exploring how computational methods can analyze complex clinical information.

Research Contribution

This project combines two important aspects of modern healthcare AI:

Prediction

Can artificial intelligence identify meaningful patterns?

Explanation

Can those predictions be understood and evaluated?

Together, these approaches create a more transparent and scientifically useful machine learning workflow.

Limitations and Responsible Interpretation

Limitations of the Study

Although the models achieved strong predictive performance, several limitations must be considered before drawing conclusions.

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1. Dataset Limitations

The study uses the ADHD-200 Consortium phenotypic dataset, which has several important characteristics:

- Participants were collected from multiple research sites
- Data collection protocols may vary between sites
- The sample size is relatively limited compared with real-world clinical populations
- The dataset may not fully represent the diversity of individuals with ADHD

Therefore, model performance on this dataset may not directly translate to other populations.

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2. Phenotypic Data Only

This project focuses exclusively on phenotypic information, including:

- Demographic characteristics
- Cognitive measurements
- Symptom-related scores

The model does not include:

- Neuroimaging data
- Genetic information
- Longitudinal patient history
- Environmental factors

ADHD is a complex disorder involving biological, psychological, and environmental influences that cannot be captured by a limited set of variables.

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3. Prediction Does Not Equal Diagnosis

A high-performing machine learning classifier does not mean that the model can diagnose ADHD.

The model:

- Identifies statistical patterns in research data
- Estimates associations between features and diagnosis
- Does not replace clinical assessment

Clinical diagnosis requires comprehensive evaluation by qualified professionals.

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4. Risk of Overfitting

Machine learning models can sometimes learn patterns specific to a dataset rather than generalizable biological relationships.

To reduce this risk, the project included:

- Train/test evaluation
- Cross-validation
- Comparison of multiple algorithms
- Explainability analysis

However, external validation on independent datasets would be necessary before real-world application.

Responsible Interpretation

The results should be understood as:

> A demonstration that machine learning can identify and explain ADHD-related patterns within a research dataset.

They should not be interpreted as:

- A clinical diagnostic system
- A replacement for medical evaluation
- Evidence of causal relationships

Main Takeaway

The strength of this project is not only predictive accuracy, but the combination of:

Machine Learning Performance + Explainable AI + Scientific Caution

Repository Structure and Reproducibility

Project Organization

The project was designed as a reproducible machine learning workflow where each stage of the analysis is separated into clear components.

The repository structure reflects the complete research pipeline:

```
ADHD-Behavioral-Prediction-AI/

│
├── data/
│   ├── raw/
│   │   └── ADHD-200 phenotypic dataset
│   │
│   └── processed/
│       └── cleaned binary classification dataset
│
├── src/
│   ├── dataset preparation scripts
│   ├── model training scripts
│   ├── benchmarking scripts
│   └── explainability analysis scripts
│
├── models/
│   └── trained machine learning models
│
├── reports/
│   ├── model performance results
│   ├── feature importance analysis
│   └── SHAP visualizations
│
└── README.md
```

Reproducible Workflow

The complete analysis can be reproduced through the following sequence:

Dataset Acquisition

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Data Cleaning & Preprocessing

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Feature Preparation

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Model Training

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Model Evaluation

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Explainability Analysis

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Research Interpretation

Implementation Approach

The project follows several principles:

Separation of Components

Data processing, model training, evaluation, and interpretation are implemented as independent steps.

This allows:

- Easier debugging
- Independent testing
- Future extension with new models or datasets

Automated Analysis

Scripts were created to automate:

- Dataset preparation
- Model comparison
- Performance evaluation
- Feature importance extraction

- SHAP analysis

This reduces manual processing and improves reproducibility.

Open Research Approach

The project is structured to allow future researchers or developers to:

- Review the methodology
- Reproduce the results
- Modify the pipeline
- Extend the analysis with additional data sources

Technologies Used

Programming Language

- Python

Machine Learning

- Scikit-learn
- XGBoost

Explainable AI

- SHAP

Data Processing

- Pandas
- NumPy

Development Environment

- GitHub Codespaces

Engineering Perspective

This repository demonstrates the integration of:

- Data engineering
- Machine learning development
- Model evaluation
- Explainable AI
- Scientific reporting

The goal was not only to create a model, but to build a complete and understandable AI research workflow.

Conclusion and Key Findings

Project Summary

This project investigated whether machine learning methods could identify ADHD-related patterns from phenotypic clinical data and whether these predictions could be interpreted using explainable artificial intelligence techniques.

The work was structured into two complementary studies:

Study 1 — Predictive Modeling

The first study evaluated whether machine learning models could classify ADHD status using demographic, cognitive, and symptom-related features.

Multiple algorithms were compared:

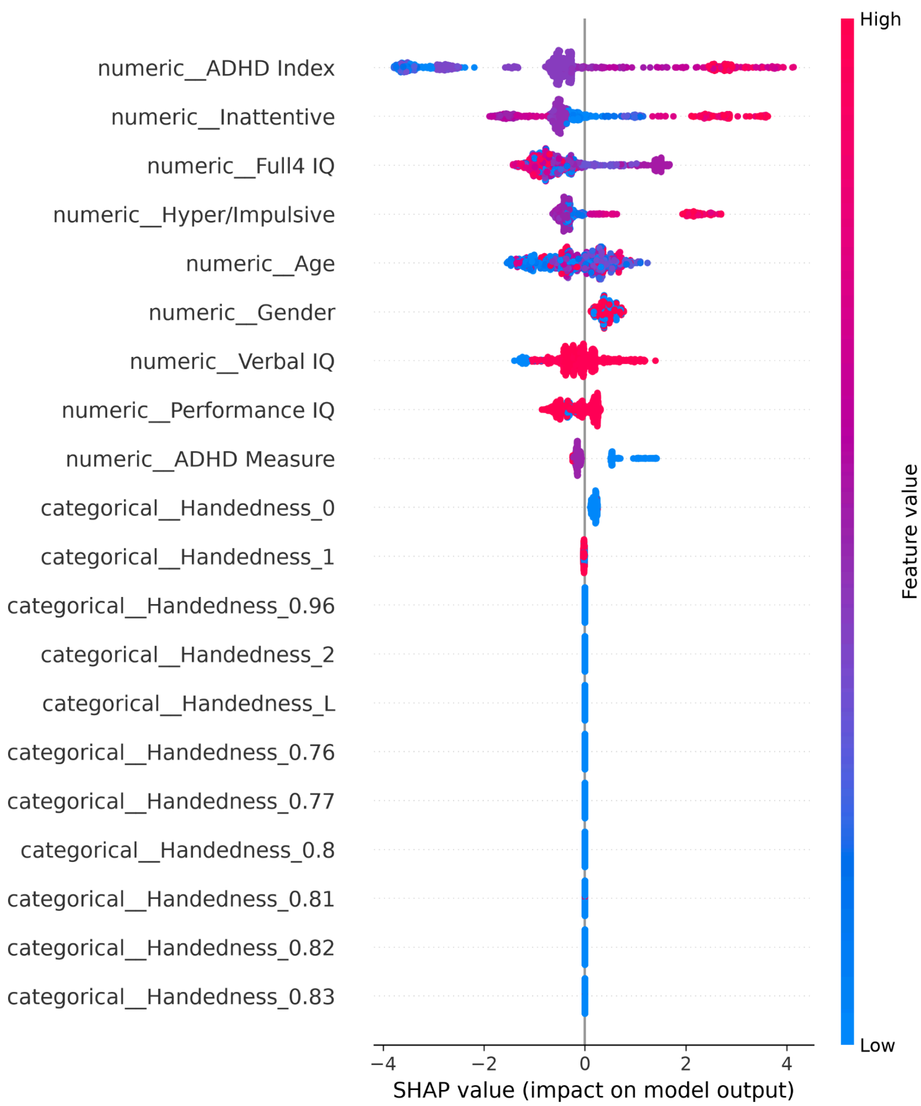
- Logistic Regression
- Support Vector Machine
- Random Forest
- XGBoost

The results showed that ensemble tree-based models achieved the strongest performance.

The final XGBoost model achieved:

- **ROC-AUC: 0.9698 during cross-validation**
- **Accuracy: 92.5% on the held-out test set**
- **ROC-AUC: 0.9356 on the test set**

Study 2 — Explainable Artificial Intelligence



The second study examined why the model produced its predictions.

Using SHAP analysis, the most influential features were identified:

1. ADHD Index
2. Inattentive symptoms
3. Full4 IQ
4. Hyperactive/Impulsive symptoms
5. Age

The strongest contributors were ADHD-related symptom measurements, demonstrating that the model relied primarily on clinically meaningful variables.

Main Findings

The project demonstrates that:

1. Phenotypic data contains predictive information

Clinical and behavioral measurements contain recognizable patterns associated with ADHD classification.

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2. Machine learning can capture complex relationships

Nonlinear models such as XGBoost outperform simpler approaches when analyzing structured clinical datasets.

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3. Explainability is essential

High predictive performance alone is insufficient.

SHAP analysis allowed investigation of whether model behavior aligned with meaningful ADHD-related characteristics.

Overall Conclusion

This project demonstrates a complete artificial intelligence workflow for healthcare-oriented research:

Clinical Dataset

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Data Processing

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Machine Learning Prediction

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Model Evaluation

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Explainable AI

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Scientific Interpretation

The results suggest that machine learning can effectively identify patterns associated with ADHD within research datasets while explainability methods provide a pathway toward more transparent and responsible AI applications.

Final Statement

> Artificial intelligence can support the analysis of complex clinical data, but meaningful healthcare applications require not only accurate predictions, but also interpretability, validation, and careful scientific evaluation.

Future Work and Project Extensions

Future Directions

This project represents an initial exploration of machine learning and explainable AI applied to ADHD-related clinical data.

Several extensions could further improve the scientific value and technical depth of the project.

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1. Integration of Additional Data Modalities

Future studies could incorporate additional sources of information:

Neuroimaging Data

The ADHD-200 Consortium also contains functional MRI data.

Future models could investigate whether combining:

- Phenotypic information
- Brain imaging features
- Behavioral measurements

improves predictive performance.

Biological and Genetic Information

Future approaches could explore:

- Genetic variants
- Biomarkers
- Physiological measurements

to investigate multimodal ADHD prediction.

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2. Advanced Machine Learning Approaches

The current project focuses on classical machine learning methods.

Future improvements could include:

- Deep learning architectures
- Neural networks
- Representation learning
- Multimodal AI models

These approaches may discover more complex patterns within large datasets.

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3. Improved Clinical Interpretability

Explainable AI analysis could be expanded through:

- Individual patient-level explanations
- Counterfactual explanations
- Feature interaction analysis
- Comparison with clinician assessments

This would provide deeper understanding of how AI reasoning relates to clinical decision-making.

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4. External Validation

A major next step would be testing the model on independent datasets.

External validation would determine whether the learned patterns generalize beyond the original ADHD-200 population.

Important validation steps:

- Testing on new research cohorts
- Evaluating performance across different populations

- Assessing robustness across clinical settings

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5. Development of Research Tools

The current pipeline could evolve into a broader research platform capable of:

- Automated dataset analysis
- Model comparison
- Explainability visualization
- Interactive clinical research dashboards

Such tools could support researchers exploring neurodevelopmental disorders using artificial intelligence.

Final Vision

The long-term goal is not to replace clinical expertise, but to develop AI systems that can:

- Analyze complex biological and behavioral information
- Reveal hidden patterns
- Support scientific discovery
- Provide transparent and interpretable results

Closing Statement

This project demonstrates the foundations of an AI research workflow:

Data → Prediction → Explanation → Scientific Understanding

By combining machine learning with explainability, computational methods can become more useful, transparent, and responsible tools for neuroscience and healthcare research.

End of Report